

Supervised Learning Setup

Spaces: Features, labels and hypotheses

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References

Machine Learning for Intelligent Systems,
CS4780/CS5780, Kilian Weinberger,
https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote01_MLsetup.html

Supervised learning

Predicting the categorical labels for unseen data based on labelled instances.

Features

Label

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia	Heart disease
1	62	70	120	80	No
2	72	90	110	70	No
3	74	80	130	70	No
4	65	120	150	90	Yes
5	67	100	140	85	Yes
6	64	110	130	90	No
7	69	150	170	100	Yes
8	66	125	145	90	?
9	74	67	110	60	?

Feature vector (4-dimensional)

Label vector

Training Data

Test Data

Supervised learning

Predicting the real-valued labels for unseen data based on labelled instances.

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia	Cholesterol Level
1	62	70	120	80	150
2	72	90	110	70	160
3	74	80	130	70	130
4	65	120	150	90	200
5	67	100	140	85	190
6	64	110	130	90	130
7	69	150	170	100	250
8	66	125	145	90	?
9	74	67	110	60	?

Rules vs. Learning

- We can write a complicated set of rules
 - Works well for a while
 - Cannot adapt well to new examples
 - Program could be reverse-engineered and circumvented
- Learn the mapping between an instance and its label using past labelled data
 - Can be retrained on new instances
 - Not easy to reverse-engineer and circumvent in all cases
 - Easier to plug the leaks

Formalizing the setup

$$D = \{(\vec{x}_1, y_1), (\vec{x}_2, y_2), \dots (\vec{x}_n, y_n)\} \subseteq X \times Y$$

Where,

- D is the data set
- X is the d -dimensional feature space ($\mathbb{R}^d$)
- $\vec{x}_i$ is the input vector of the i th sample
- Y is the label space

The data points are drawn from an unknown distribution P

$$(\vec{x}_i, y_i) \sim P(x, y)$$

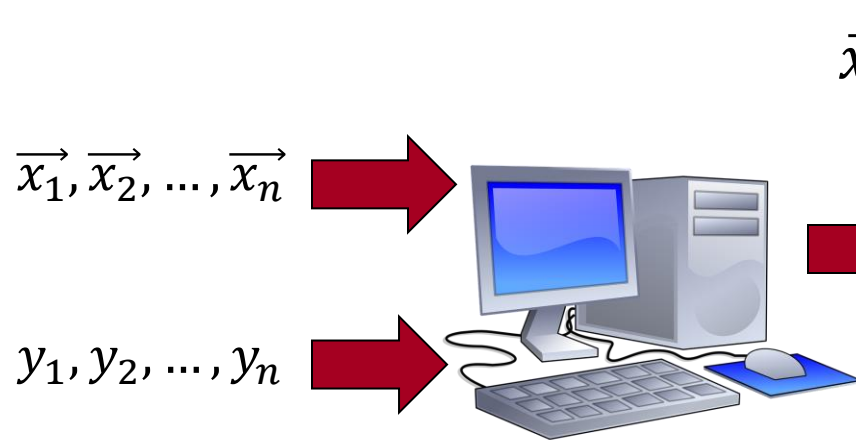
We want to learn a function $h \in H$, such that for a new instance $(\vec{x}, y) \sim P$,

$$h(x) = y$$

With a high probability or at least, $h(x) \approx y$.

Training and testing

Training



Testing



Evaluation

Compare $h(\vec{x})$ and y

Feature Space

- **Categorical**
 - **Nominal:** No natural order
 - E.g., Gender (M,F), ethnicity, hair color, weather (rainy, sunny, cloudy), blood type,...
 - Operations: Frequencies, mode, but not medians
 - **Ordinal:** A specific natural order
 - E.g., User experience (1 – 5, very unsatisfied, unsatisfied, neutral, satisfied, very satisfied), rainfall (dry, damp, wet, torrential), socio economic status (“low income”, “middle income”, “high income”), education level (“high school”, “BS”, “MS”, “PhD”),...
 - Operations: Median
 - **Differences** are not meaningful

Feature Space

- **Numerical** for quantitative data
 - Differences between values have consistent meaning
 - Finite or infinite scales
 - Add/subtract, mean/median/mode, variance
 - **Interval:** No true origin (zero). Can fall below zero.
 - E.g., temperature (in F or C), time (e.g., Gregorian or Islamic calendar), test scores, IQ scores, SAT scores...
 - A temperature of 0 C does not mean “no heat”
 - Differences are meaningful, ratios are not
 - 100 degrees C is not twice as hot as 50 C
 - **Ratio:** There is a true origin (zero). Never falls below zero.
 - E.g., Height, weight, age, temperature in K, length
 - Height = 0 means “no height”
 - An age of 50 is twice the age of 25
 - Ratios are meaningful
 - **Discrete** and **continuous**
 - E.g., visits to the dentists vs. height

Feature Space

- Arrays, embeddings, lists,...

Feature Space

- Student data (e.g., for predicting grades)

$$\mathbf{x}_i = (x_i^1, x_i^2, \dots, x_i^d)$$

- Where each x_i^j is the value of the j th feature for student i .
- Examples of features, x^j
 - Scores in assignments, quizzes, exams
 - Educational records, grades in previous courses
 - Rankings of previous educational institutes
 - Interaction with online tools? Missed instruments?
- Small number of features and relatively few features with 0 values: **Dense vectors**

Feature Space

- Image data (e.g., for facial recognition)

$$\mathbf{x}_i = (x_i^1, x_i^2, \dots, x_i^{3d})$$

- Where each group of three consecutive x^j s represent the red, green and blue values of each pixel in image i .
- In a 6-megapixel images, there are 6 million pixels and 18 million features
- Almost all pixels have non-zero values and the feature vector is mostly **dense**

x_i^1, x_i^2, x_i^3	x_i^4, x_i^5, x_i^6	x_i^7, x_i^8, x_i^9	...	
...	200, 89, 55	213, 22, 55	255, 7, 0	...
				$x_i^{58}, x_i^{59}, x_i^{60}$

Feature Space

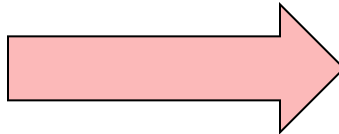
- Textual data (e.g. for authorship attribution)

$$x_i = (x_i^1, x_i^2, \dots x_i^d)$$

- Where x_i^k or $TF(k)$ is the number of occurrences of the k th dictionary word in document i : Bag-of-Word features
- English has nearly 600,000 word-forms (273,000 headwords)
- Most of the feature values are zero: Sparse vectors

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

<https://www.lipsum.com/>



a	0
abandon	0
ability	0
able	2
about	3
above	0
abroad	0
...	
Probably	0
product	1
production	0
profession	0
professional	0
professor	2
...	
youth	0
zest	0
zone	0
zoo	0

One-hot vectors

- A single high value among zeros in the vector
- One-cold vectors (all 1's with one 0)

Binary	One-hot
000	00000001
001	00000010
010	00000100
011	00001000
100	00010000
101	00100000
110	01000000
111	10000000

Label Space

- **Binary (one-of-two) – Binary classification**
 - Sentiment: positive / negative
 - Email: Spam / Not Spam
 - Online Transactions: Fraudulent (Yes / No)
 - Tumor: Malignant / Benign
 - $y \in \{0,1\}$ e.g. 0: Negative class, 1: Positive class
 - $y \in \{-1,1\}$ e.g. -1: Negative class, 1: Positive class
- **Multi-class (one-of-many, many-of-many problems) – Multi-class classification**
 - Sentiment: Positive / negative / neutral
 - Emotion: Happy, Sad, Surprised, Angry,...
 - Part-of-Speech tag: Noun / verb / adjective / adverb / ...
 - Recognize a word: One of $|V|$ tags
 - $y \in \{0,1,2,3, \dots\}$ e.g. 0: Happy, 1: Sad, 2, Angry,...
- **Real-valued – Regression**
 - Temperature, height, age, length, weight, duration, price...

Hypothesis Space

- The hypothesis h is sampled from a hypothesis space H .

$$h \in H$$

- H can be thought of to contain classes of hypotheses which share sets of assumptions like
 - Decisions tree
 - Perceptron
 - Neural networks
 - Support Vector Machines
- For example, $h \in H$ for H : Decision trees, would be instances of decisions trees of different height, arity, thresholds, etc.

So, how do we choose our h ?

- Randomly?
- Exhaustively?

How do we evaluate h ?

For more details please visit
<http://aghaaliraza.com>



Thank you!